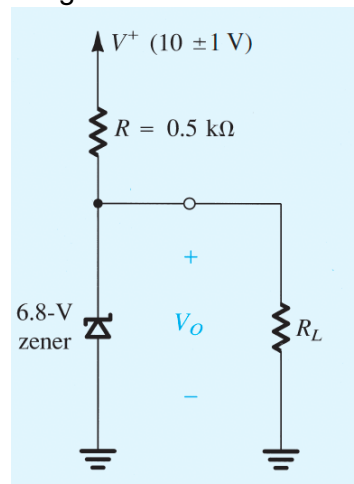
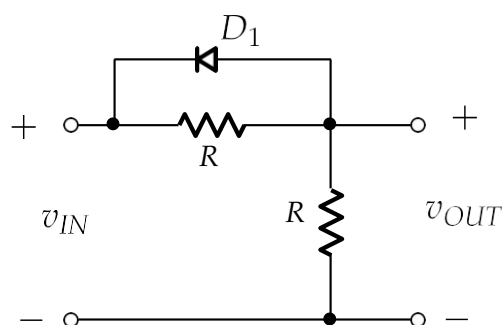


Homework 8 Topic: Diodes II + BJTs

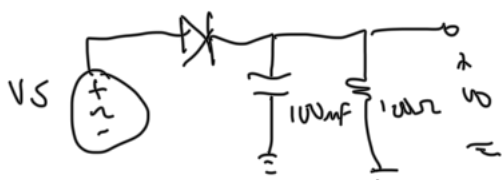
- Consider a half-wave peak rectifier fed with a voltage v_s having a triangular waveform with 24-V peak-to-peak amplitude, zero average, and 1-kHz frequency. Assume that the diode has a 0.7-V drop when conducting. Let the load resistance $R = 100\Omega$ and the filter capacitor $C = 100\mu\text{F}$. Find the average dc output voltage, the time interval during which the diode conducts, the average diode current during conduction, and the maximum diode current.
- Zener Diode Regulator: For this problem you will need the data sheet for a Zener Diode, which you can find your own or use the one Fairchild Semiconductor that is provided on Canvas. Note that the data sheet provides values for V_Z , which is measured at the current I_Z , as well as for the series resistance. We want to design a 12 V regulator using a reverse-bias Zener diode similar to the circuit below the load of $2.2\text{ k}\Omega$. For this design problem, the supply voltage V^+ is $25 \pm 2.5\text{ V}$ (not $10 \pm 1\text{ V}$ as shown in the figure below).
 - Choose one Zener diode to regulate the load voltage to approximately 12 V. Based on the information in the data sheet, find V_{Z0} for this diode.
 - If the Zener current is 25 mA, find V_Z . Then find the current through R , and from that determine a value for R . Use the nominal voltage for the power supply (i.e., 25 V).
 - Find the line regulation.



- For the circuit below using the constant voltage drop model with $V_{D0} = 0.7\text{ V}$ and $r_D = 0$:
 - Write an expression for v_{OUT} in terms of v_{IN} .
 - Sketch a graph of v_{OUT} vs. v_{IN} .



1)



$$V_Z = 0.7 \text{ V}$$

$$V_S = 2 \text{ V}_{pkpk}$$

$$1 \text{ kHz}$$

$$T = 10^{-3} \text{ s}$$

$$\begin{aligned} V_{avg, DC} &= 10.76 \text{ [V]} \\ \Delta t &= 0.0225 \text{ [ms]} \\ i_{Z, avg} &= 4.9076 \text{ [A]} \\ i_{Z, max} &= 4.913 \text{ [A]} \end{aligned}$$

$$V_{O, pk} = V_{S, pkpk} - 0.7 \text{ V} = \frac{2 \text{ V}}{2} - 0.7 = 1.3 \text{ [V]}$$

$$v_O = 1.3 e^{-t/\tau} \Rightarrow v_O @ \tau = T = 10.22 \text{ [ms]} \text{ max decay}$$

$$\Delta V_O = 1.3 \cdot 10.22 = 1.08$$

$$I_L = \frac{V_{avg}}{R_L} = \frac{10.76}{100} = 0.1076 \text{ A} \quad I_{avg} = I_L + I_D = 4.8 + 0.1076 = 4.9076 \text{ A}$$

$$i_C \Delta t = C \Delta V$$

$$i_C = \frac{100 \times 10^{-9} \cdot 1.08}{22.5 \times 10^{-9}} = 4.8 \text{ A} \quad (I_{pk} = C \frac{\Delta V}{\Delta t} \Rightarrow 100 \cdot \frac{1.08 \text{ V}}{10^{-3}} = 4.8 \text{ A}) \quad i_{Z, max} = i_{C, pk} + i_{L, pk} = 4.8 + 0.1076 = 4.913 \text{ A}$$

$$i_{L, pk} = \frac{V_O}{R_L} = \frac{1.3}{100} = 0.013$$

2)



$$Z_D = \text{MM5221B} \quad r_{D, on} = 30 \Omega \quad V_{nom} = 2.4 \text{ [V]}$$

$$V_{Z0} = V_Z - I_Z r_Z = 12 - 30 \cdot 20 \cdot 10^{-3} = 11.4 \text{ V}$$

$$I_Z = 29 \text{ mA} \quad V_Z = V_{Z0} + I_Z r_Z = 11.4 + 30 \cdot 29 \cdot 10^{-3} = 12.15 \text{ V}$$

$$I_L = I_Z + I_{R_L} = 2.5 - 12.15 = 2.5 + \frac{12.15}{2.2 \cdot 10^3} \Rightarrow R_L = \frac{12.85}{30.52 \cdot 10^{-3}} = 421$$

$$\frac{V_S - V_Z}{R} = I_L + \frac{V_Z}{R_L}$$

$$LR = \Delta V_S \frac{R_Z}{R_1 R_2} = \frac{2.5 \text{ V} \cdot 30}{421 \cdot 30} = 0.166 \frac{\text{V}}{\text{V}}$$

$$\begin{aligned} V_{Z0} &= 11.4 \text{ [V]} \\ V_Z &= 12.15 \text{ [V]} \\ R &= 421 \Omega \end{aligned}$$

$$LR = 166 \frac{\text{mV}}{\text{V}}$$

3) $V_{D0} = 0.7 \text{ V} \quad r_D = 0$

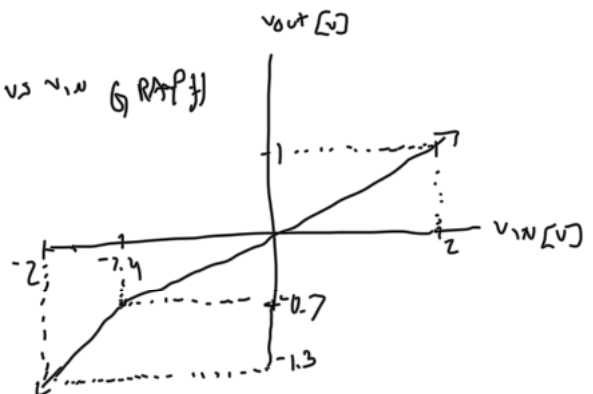
a) $V_{out} = -v_{in}$

$$\frac{V_{out}}{R} = \frac{v_{in} - (V_{out} + 0.7)}{R}$$

$$V_Z = V_{out} + v_{in} = 3 \text{ V out}$$

$$V_{out} = \begin{cases} v_{in} + 0.7, & v_{in} \leq -1.4 \text{ [V]} \\ v_{in}/2, & v_{in} > -1.4 \text{ [V]} \end{cases}$$

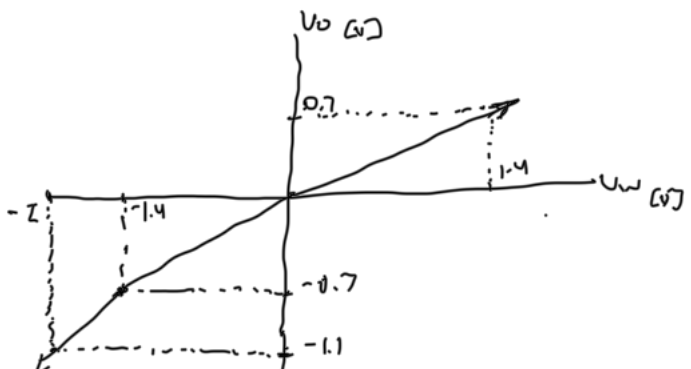
b) V_{out} vs v_{in} (GRAPH)



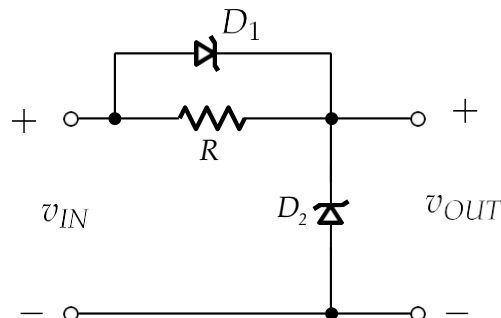
4) $V_{out} = \begin{cases} \frac{2v_{in} + 0.7}{3}, & v_{in} \leq -1.4 \text{ [V]} \\ \frac{v_{in}}{2}, & v_{in} > -1.4 \text{ [V]} \end{cases}$

$$\frac{V_D}{R} = \frac{v_{in} - V_D}{R} - \frac{V_{out} - 0.7}{R}$$

$$\frac{2V_D}{R} = \frac{(2v_{in} + 0.7)}{3R}$$



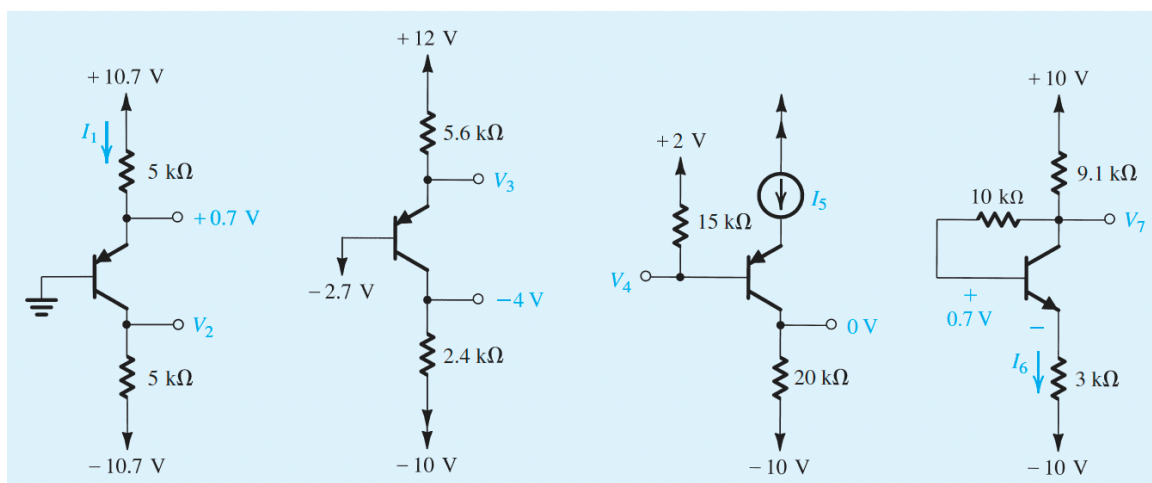
4. Repeat problem 3 with $r_D = R$.
5. For the circuit below using the constant voltage drop model with $V_{D0} = 0.7\text{ V}$, $V_{Z0} = 1.4\text{ V}$, $r_D = r_z = 0$:
 - a. Write an expression for v_{OUT} in terms of v_{IN} .
 - b. Sketch a graph of v_{OUT} vs. v_{IN} .



6. Measurements of V_{BE} and two terminal currents taken on a number of npn transistors operating in the active mode are tabulated below. For each, calculate the missing current value as well as α , β , and I_S as indicated by the table.

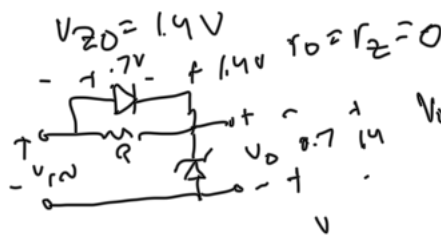
Transistor	a	b	c	d	e
V_{BE} (mV)	700	690	580	780	820
I_C (mA)	1.000	1.000		10.10	
I_B (μA)	10		5	120	1050
I_E (mA)		1.020	0.235		75.00
α					
β					
I_S					

7. For the circuits below, assume that the transistors have very large β . Some measurements have been made on these circuits, with the results indicated in the figure. Find the values of the other labeled voltages and currents.



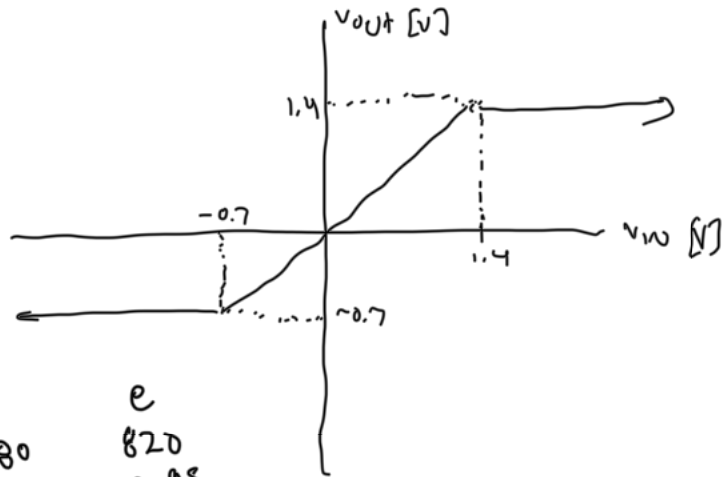
5)

$$V_{DD} = 0.7V$$



$$V_{out} = \begin{cases} V_{in} & V_{in} \leq -0.7V \\ -0.7V & -0.7V < V_{in} < 1.4V \\ V_{in} - 0.7V & V_{in} \geq 1.4V \end{cases}$$

$$\begin{aligned} V_{in} &\leq -0.7V \\ -0.7V &< V_{in} < 1.4V \\ V_{in} &\geq 1.4V \end{aligned}$$



b)

	a	b	c	d	e
$V_{BE} (mV)$	700	690	580	780	820
$I_C (mA)$	1.000	1.000	0.23	10.10	13.95
$I_B (mA)$	10	20	5	120	1050
$I_E (mA)$	1.01	1.020	0.235	10.22	15.00
α	.99	.98	.979	.988	.986
β	100	50	46	84.167	70.43
I_S	6.91×10^{-16}	1.0315×10^{-15}	1.932×10^{-15}	2.847×10^{-16}	4.208×10^{-16}

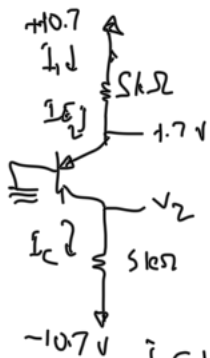
$$\alpha = \frac{\beta}{\beta + 1} \quad \beta = \frac{\alpha}{1 - \alpha}$$

$$\begin{aligned} I_E &= I_B + I_C \\ I_B &= \frac{I_C}{\beta} \\ I_E &= \frac{I_C}{\alpha} \end{aligned}$$

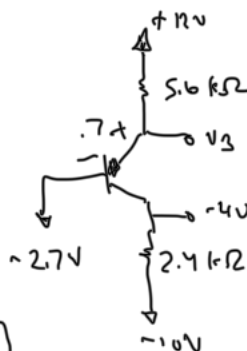
$$I_S = \frac{I_C}{e} \frac{V_{BE}}{V_T}$$

$$T = 300K \quad V_T = 25mV$$

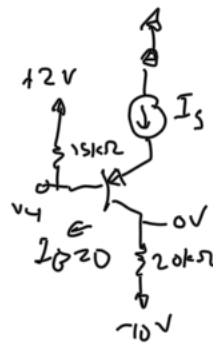
7) $\beta \gg \alpha$



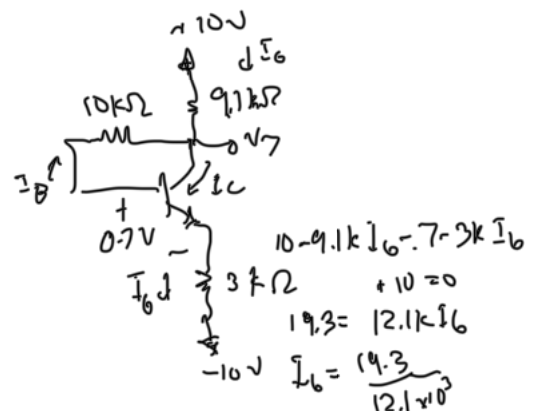
$$\begin{aligned} I_1 &= \frac{10.7 - 0.7}{5k\Omega} = 2mA \\ I_C &= I_E = I_1 = 2mA \\ V_2 &= 10.7 - 5k\Omega \cdot 2mA = 0.7V \end{aligned}$$



$$\begin{aligned} V_3 &= 10.7 - 5.6k\Omega \cdot 2mA = -2.0V \end{aligned}$$



$$\begin{aligned} I_S &= I_E = I_C \\ I_S &= \frac{12 - 0.7}{20k\Omega} = 0.5mA \\ V_4 &= 12 - 20k\Omega \cdot 0.5mA = 2V \end{aligned}$$



$$\begin{aligned} I_6 &= \frac{10 - 0.7}{10k\Omega} = 0.93mA \\ V_7 &= 10 - 9.1k\Omega \cdot 0.93mA = -4.51V \end{aligned}$$